Dataset 1: Spam Email / SMS Dataset

Purpose:

Used for **binary text classification** to identify spam messages.

Key Details

- **Type:** Text dataset (SMS / Email messages)
- **Total samples:** ~5,157 messages
- **Number of classes: 2**

  - ham → Not spam
  - spam → Spam

Class distribution:

- Ham = 87%
- Spam = 13%

LINK: https://www.kaggle.com/datasets/mfaisalqureshi/spam-email

Dataset 2: MNIST Handwritten Digit Dataset

Purpose:

Used for **multiclass image classification** to recognize handwritten digits.

Key Details

- **Type:** Image dataset (grayscale)
- **Total samples: 70,000 images**
  - 60,000 training
  - 10,000 testing
- **Number of classes: 10**
  - Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- **Image size:** 28 × 28 pixels
- **Features:**
  - Pixel intensity values (0–255)
  - Flattened into 784 features for classical ML

LINK: https://www.kaggle.com/datasets/oddrationale/mnist-in-csv

Project - Predicting House Prices (California Dataset)

- The California Housing dataset contains **20,640 records** representing different districts in California.
- It includes **8 numerical features** such as median income, average rooms, population, and location (latitude & longitude).
- The target variable is **MedHouseVal**, which represents the median house value.
- The dataset is clean and suitable for **regression analysis**, with no missing values.
- It is widely used for practicing **machine learning and predictive modeling tasks**.

Project - Iris Flower Classification:

- The Iris dataset is a supervised classification dataset containing **150 flower samples**.
- It includes **3 classes**:
  Setosa
  Versicolor
  Virginica (50 samples each)
- Each sample has **4 numerical features**: sepal length, sepal width, petal length, and petal width.
- The dataset is **clean and balanced**, with no missing values.
- It is widely used for learning and testing classification algorithms.